WRITTEN EXAM

Course Name: Modellering och reglering / Modeling and Control	**Course Code:** R0004E
Number of examination hours: 5	**Date:** 2024-08-28

Allowed aids:

- non-programmed calculator
- ”formelsamling i reglerteknik” document
- an A4-sheet (both sides) with own notes that are submitted with the solution sheets

On duty teacher (complete telephone number):	**On duty teacher (complete telephone number):**
Dina Shona Laila, +46 76 049 2718	Arman Sehatnia, +46 72 748 9994

Grade scale:	**Number of questions and total score:**
3 – 12 points, 4 – 16 points, 5 – 20 points	7 questions, 25 points

Other information:

General instructions

Check that you have received all the exams/questions.

All new answers begin on a separate page. Do not write on the back page.

Use the provided semi-log paper to draw the Bode plot.

Print, write clearly, and do not use red pen.

After examination

Examination results will be announced within three weeks from the examination date and no later than two weeks before next re-examination.

The exam result can be seen at My LTU – Ladok for students.

Your corrected exam is scanned and visible on My LTU – Graded exams.

Information to the printer

Project number:	**Number of copies**	**Number of pages**
341980	73	4
Other information: Please attach the Appendices on each copy		

1. (a) Hardware implementation of a controller can be realized for instance using a microcontroller and a Programmable Logic Controller (PLC). Please explain at least two differences of the two types of controller hardware. (1)

   (b) Provide one example each, of control system applications where

      i) microcontroller

      ii) PLC

      is used. (1)

   (c) Draw the block diagram of a feedback control system. Then list and explain the components that are necessary in the implementation of a feedback control system, which can be excluded from an open-loop control system. (1)

Total for Question 1: 3 points

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2. Given an RLC circuit as shown in Figure 2, with the input $v(t)$ and the output $v_C(t)$.

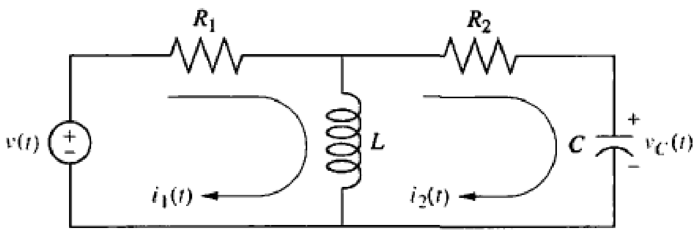

Figure 1: An RLC circuit

The dynamic equation of this circuit can be derived using the Ohm Law and the Kirchoff Laws, resulting in the following set of equations in Laplace domain:

$$(R_1 + Ls)I_1(s) - LsI_2(s) = V(s)$$
$$(R_2 + Ls + \frac{1}{Cs})I_2(s) - LsI_1(s) = 0$$
$$V_c(s) = \frac{1}{Cs}I_2(s)$$

   (a) Obtain the transfer function $\frac{V_C(s)}{V(s)}$ of the circuit, given the components $R_1 = 1\Omega$, $R_2 = 1\Omega$, $C = 1$F and $L = 1$H. (1)

   (b) Find the Laplace inverse of $V_C(s)$ if the input is a unit step, i.e. $V(s) = \frac{1}{s}$. (1)

   (c) Find the steady-state value (final value) of $v_C(t)$ with the input $v(t)$ a unit step, by applying the Final Value Theorem to (a). (0.5)

   (d) Repeat the question (c), by evaluating the Laplace inverse obtained in (b), then compare the answer with your answer in (c). (0.5)

Total for Question 2: 3 points

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3. A shaker table system is modeled using mass-spring-damper elements, as shown in Figure 2. The mass of the table is M, the mass of the camera is m, the spring stiffness constant is K_s, the damping constant is B. The measured quantity is the displacement of the table and the input to the system is the force $f_i(t)$. Using Newton's second law, the dynamics of the system can be represented by the second-order differential equation

$$a\ddot{x}(t) + b\dot{x}(t) + cx(t) = u(t)$$

where $x(t)$ is the displacement and $u(t)$ is the input force from the coil.

(a) Substitute the parameters a, b and c with the physical parameters of the system and determine the input and the output of the system. (1)

(b) Derive the state-space model of the system, by finding the matrices A, B, C and D, and write the state-space representation of the system. (2)

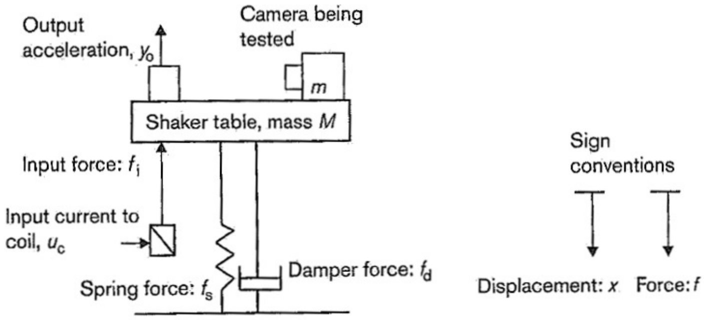

Figure 2: Schematic diagram of a shaker table and forces acting on it

(c) Given $m = 250$ g, $K_s = 1.1 \times 10^7$ N/m and $B = 1100$ N/ms, determine the range of values of M such that the system is stable. Provide explanation to your answer. (1)

Total for Question 3: 4 points

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4. A control system has a state-space model represented by:

$$\dot{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u; \qquad y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x$$

(a) Determine the stability of the system and mention the type of the stability. (0.5)

(b) Show that the equivalent transfer function of the system is

$$G(s) = \frac{3s + 5}{s^2 + 4s + 6}$$ (1)

(c) Determine the closed loop transfer function $T(s)$ with unity feedback. (0.5)

(d) Find the steady state error of each the open-loop system $G(s)$ and the closed-loop system $T(s)$, with respect to a unit step input. Explain your answer by relating the steady-state errors that you find, to the type of the system. (1.5)

(e) If the steady-state errors are not zero, explain how to eliminate the steady-state error in the closed-loop system. (0.5)

Total for Question 4: 4 points

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5. Consider the position control system in Figure 3, in which K is an adjustable gain parameter and

$$G(s) = \frac{(s + 3)}{(s + 2)(s^2 + 2s + 25)}$$

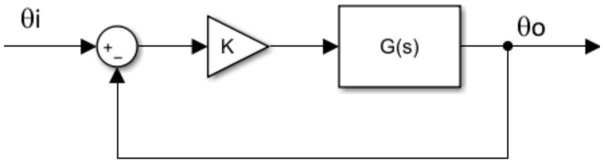

Figure 3: Block diagram of a position control system.

(a) On the semilog graph paper(s) provided, sketch the Bode plot, both the magnitude and phase plots, of the open-loop transfer function $KG(s)$, with $K = 2$. Include the sketch of the Bode plot of each term. (3)

(b) Find the gain margin (GM) and the phase margin (PM) of the Bode plot. (0.5)

(c) Explain whether the system is stable or not, based on the value of GM and PM that you obtain in (b). (0.5)

Total for Question 5: 4 points

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6. The open-loop transfer function of a system is given by

$$G(s) = \frac{K(s^2 - 8s + 15)}{s^2 + 2s + 8}$$

For $K = 1$, the Nyquist plot of $G(s)$ is shown in Figure 4.

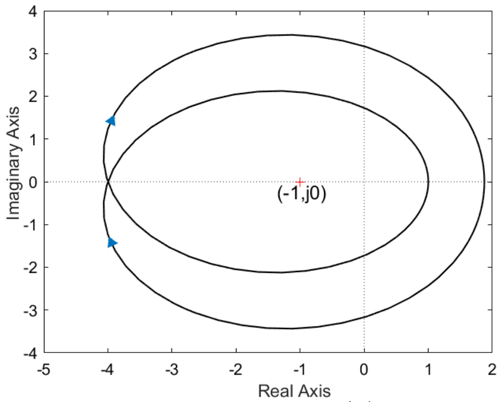

Figure 4: Nyquist plot for $G(s)$ with $K = 1$.

(a) Considering positive frequency, determine where the Nyquist plot starts (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = 0$) and where the plot ends (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = \infty$). (1.5)

(b) Find the value of N, P and Z, and determine the stability of the closed-loop system using the Nyquist criterion. (1.5)

Total for Question 6: 3 points

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7. The open-loop transfer function of a 2^{nd} order plant is given by

$$G(s) = \frac{6}{s^2 + 2s + 6}$$

(a) A PD controller $(K_p + K_d s)$ is added in series with the plant. Find the closed-loop transfer function of the system and write the closed-loop characteristic equation in terms of K_p and K_d. (1)

(b) Find the 2^{nd} order system parameters in terms of K_p and K_d, for the closed-loop transfer function. (0.5)

(c) Determine the value of the proportional gain K_p and the differential gain K_d, such that the 2^{nd} order parameters of the closed loop system has the values $\omega_n = 8$ and $\zeta = 0.8$. (2)

(d) Explain what the expected effect of the use of the PD controller in this system. (0.5)

Total for Question 7: 4 points

END

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KURSKOD / COURSE CODE	PROV / TEST CODE
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YYYYMMDD-XXXX

[redacted]

KURSBENÄMNING / COURSE NAME

Modellering och reglering

PROVBENÄMNING / TEST NAME

Tentamen

TENTAMENSDATUM / EXAMINATION DATE

2024-08-28

TENTAMENSORT/CITY (för distansstudenter / for off campus students only)

PROGRAM	INSKR.ÅR/YEAR	ANTAL SIDOR / NO. OF PAGES
TCDAA	2020	07

Skanningsblad/Scanning Sheet

Behandlat uppgift nr (sätt x) / Mark the questions you answered with an X

Lärarens anteckningar / Teacher's notes

1	X	2.3	
2	X	3	
3	X	4	
4	X	3.5	
5	X	3.4	
6		0	
7	X	3	
8			
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11			
12			
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18			

Poängsumma Points	19.2	Betyg Grade	4

35065

Tentamensomslag skall alltid inlämnas även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2024-08-05 kl. 08:25:04

Namn:	Personnummer:	Sidnr:
[redacted]	[redacted]	1

Uppgift nr: 1

Poäng: 2.3

Lärarens anteckning:

not always weak

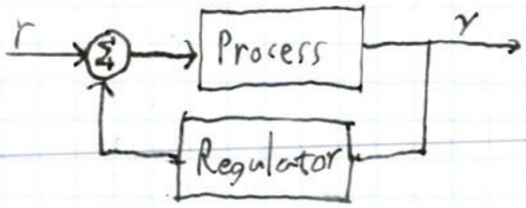
a. 1) A microcontroller is simply a small and quite weak computer that can be used for many purpouses while the PLC is specialized for logic.

2) PLCs are more suitable for large-scale processes than microcontrollers. ✓

b. i) A washing machine. ✓ 1

ii) A paper mill. ✓ 1

c.



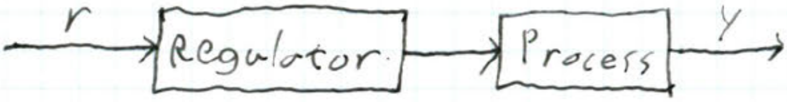
no need to split it is just a split 1.

For an open-loop you would have no splitting of the output from the process (the feedback) which is necessary in order to respond to the process. You also wouldn't have the summation (Σ) that calculates the error e.

comparator

0.3

An open-loop system would look more like this:



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Namn: [redacted] Personnummer: Sidnr: 2

Uppgift nr: 2

Poäng: 3

Lärarens anteckning:

a. $\frac{V_c}{V} = \frac{\frac{1}{Cs} I_2}{(R_1+Ls)I_1 - LsI_2} = \frac{\frac{1}{s}I_2}{(1+s)I_1 - sI_2}$ (1)

$V_c = \frac{I_2}{Cs} \Leftrightarrow I_2 = sV_c$ (2) $\quad (R_2+Ls+\frac{1}{Cs})I_2 - LsI_1 = 0$ (3)

(2) in (3): $(1+s+\frac{1}{s})sV_c - sI_1 = 0 \Leftrightarrow I_1 = (1+s+\frac{1}{s})V_c$ (4)

(2,4) in (1): $\frac{\frac{1}{s} sV_c}{(1+s)(1+s+\frac{1}{s})V_c - s^2V_c} = \frac{V_c}{V} \Leftrightarrow \frac{V_c}{V} = \frac{1}{1+s+\frac{1}{s}+s+s^2+1-s^2} =$

$= \frac{1}{2s+\frac{1}{s}+2} = \frac{1}{\frac{2s^2+2s+1}{s}} = \frac{s}{2s^2+2s+1}$ Answer: $\frac{V_c}{V} = \frac{s}{2s^2+2s+1}$ ✓ 1

b. $V_c(s) = \frac{1}{s} \cdot \frac{s}{2s^2+2s+1} = \frac{1}{2s^2+2s+1} = \frac{1}{2} \cdot \frac{1}{s^2+s+0,5}$ w_0 2, w ×

From Laplace transforms: $\frac{1}{s^2+2\zeta w_0 s + w_0^2}$ is $\frac{e^{-\zeta w_0 t}}{w_0\sqrt{1-\zeta^2}}\sin(w_0\sqrt{1-\zeta^2}$

if $0<\zeta<1$ From the constants we get $w_0^2 = 0,5$ & $2w_0\zeta = 1$ 1

$w_0 = \sqrt{0,5}$ & $\zeta = \frac{1}{2\sqrt{0,5}}$ therefore we get:

$V_c(t) = e^{-\frac{1}{2}t} \cdot \sin(0,5t)$ ← Answer ✓

0.5

c. FVT: $\lim_{s\to 0} sV_c(s)V(s) = \lim_{s\to 0} s \cdot \frac{s}{2s^2+2s+1} \cdot \frac{1}{s} = \frac{0}{1} = 0$ ✓

d. $V_c(t) = 2e^{-\frac{1}{2}t} \cdot \sin(0,5t)$ $t \to \infty \Rightarrow e^{-\frac{1}{2}t} \to 0 \Rightarrow V_c(t) \to 0$ 0.5

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Uppgift nr: 3

Poäng: 4

Lärarens anteckning:

a. According to $F=ma$ a should be $(m+M)$ since that is the total mass of the table and $u(t)$ is the force while $\ddot{x}$ is the acceleration.
b should be B since damping depends on velocity $(\dot{x})$.
Finally c should be K_s since spring force depends on displacment. So: $(M+m)\ddot{x}(t) + B\dot{x}(t) + K_s x(t) = u(t)$ ✓ F_i
The output $y = x$ 1

b. $x_1 = x$, $x_2 = \dot{x}$ | $\dot{x}_1 = x_2$, $\dot{x}_2 = \ddot{x} = \dfrac{-K_s x_1 - B x_2 + u}{(M+m)}$, $y = x_1$ |

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ \frac{-K_s}{(M+m)} & \frac{-B}{(M+m)} \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{(M+m)} \end{bmatrix}}_{B} u \checkmark$$

$$y = \underbrace{[1 \quad 0]}_{C} x + \underbrace{0}_{D} \cdot u$$

2

c. Stability only depends on A. The system is stable as long as the real parts of A's eigenvalues are negative. 1

$$\det(sI - A) = \det\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ \frac{-1{,}1\cdot 10^7}{(M+0{,}25)} & \frac{-1100}{(M+0{,}25)} \end{bmatrix}\right) = \checkmark$$

✓

$$= \det\left(\begin{bmatrix} s & -1 \\ \frac{1{,}1\cdot 10^7}{(M+0{,}25)} & s + \frac{1100}{(M+0{,}25)} \end{bmatrix}\right) = s^2 + \frac{1100 s}{(M+0{,}25)} + \frac{1{,}1\cdot 10^7}{(M+0{,}25)}$$

$$PQ: s = \frac{-550}{(M+0{,}25)} \pm \sqrt{\left(\frac{550}{(M+0{,}25)}\right)^2 - \frac{1{,}1\cdot 10^7}{(M+0{,}25)}} = \frac{-550}{(M+0{,}25)} \pm \sqrt{\frac{302500}{(M+0{,}25)^2} - \frac{1{,}1\cdot 10^7}{(M+0{,}25)}}$$

We want $\dfrac{550}{(M+0{,}25)} > \sqrt{\dfrac{302500}{(M+0{,}25)^2} - \dfrac{1{,}1\cdot 10^7}{(M+0{,}25)}}$ in order to have a negative real part (without negative mass)

Since $\dfrac{302500}{(M+0{,}25)^2} > \dfrac{302500}{(M+0{,}25)^2} - \dfrac{1{,}1\cdot 10^7}{(M+0{,}25)}$ for all M it is probable that my physical parameters are incorrect, but the principle holds.

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Namn:	Personnummer:	Sidnr:
[redacted]		4

Uppgift nr: 4

Poäng: 3.5

Lärarens anteckning:

a) 0.5

b. 1

0.5

~~0.8~~ 1

0.5

a. $\det(sI - A) = \det\left(\begin{bmatrix} s+4 & 1,5 \\ -4 & s \end{bmatrix}\right) = s^2 + 4s + 6$ pq: $s = -2 \pm \sqrt{4-6}$

$s = -2 \pm \sqrt{2}\, i$ the system is asymptotically stable since all real values are solidly in the LHP

b. $G = C(sI - A)^{-1} B + D = [1,5 \;\; 0,625] \frac{1}{s^2+4s+6} \begin{bmatrix} s & -1,5 \\ 4 & s+4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} =$

$= [1,5 \;\; 0,625] \frac{1}{s^2+4s+6} \begin{bmatrix} 2s \\ 8 \end{bmatrix} = \frac{3s+5}{s^2+4s+6} = G(s)$, Shown!

c. Unity feedback: $\frac{G(s)}{1+G(s)} = \frac{\frac{3s+5}{s^2+4s+6}}{1 + \frac{3s+5}{s^2+4s+6}} = \frac{3s+5}{s^2+7s+11} = T(s)$

d. FVT: $\lim_{s \to 0} s G(s) \cdot \frac{1}{s} = \lim_{s \to 0} \frac{3s+5}{s^2+4s+6} = \frac{5}{6} = e_{ss}$. ← not error $E = 1 - \frac{5}{6} = \frac{1}{6}$ This is expected since $G(s)$ is a type 0 system, which produce steady-state errors on unit step input.

→ not error $E = 1 - \frac{5}{11} = \frac{6}{11}$

FVT: $\lim_{s \to 0} s T(s) \cdot \frac{1}{s} = \lim_{s \to 0} \frac{3s+5}{s^2+7s+11} = \frac{5}{11} = e_{ss}$ While the error is smaller we still get an error since $T(s)$ is also a Type 0 system.

e. We add an integrator $\frac{1}{s}$ to the system.

For example:

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Uppgift nr: 5

Poäng: 3,4

Lärarens anteckning:

a. See included Bode diagram.

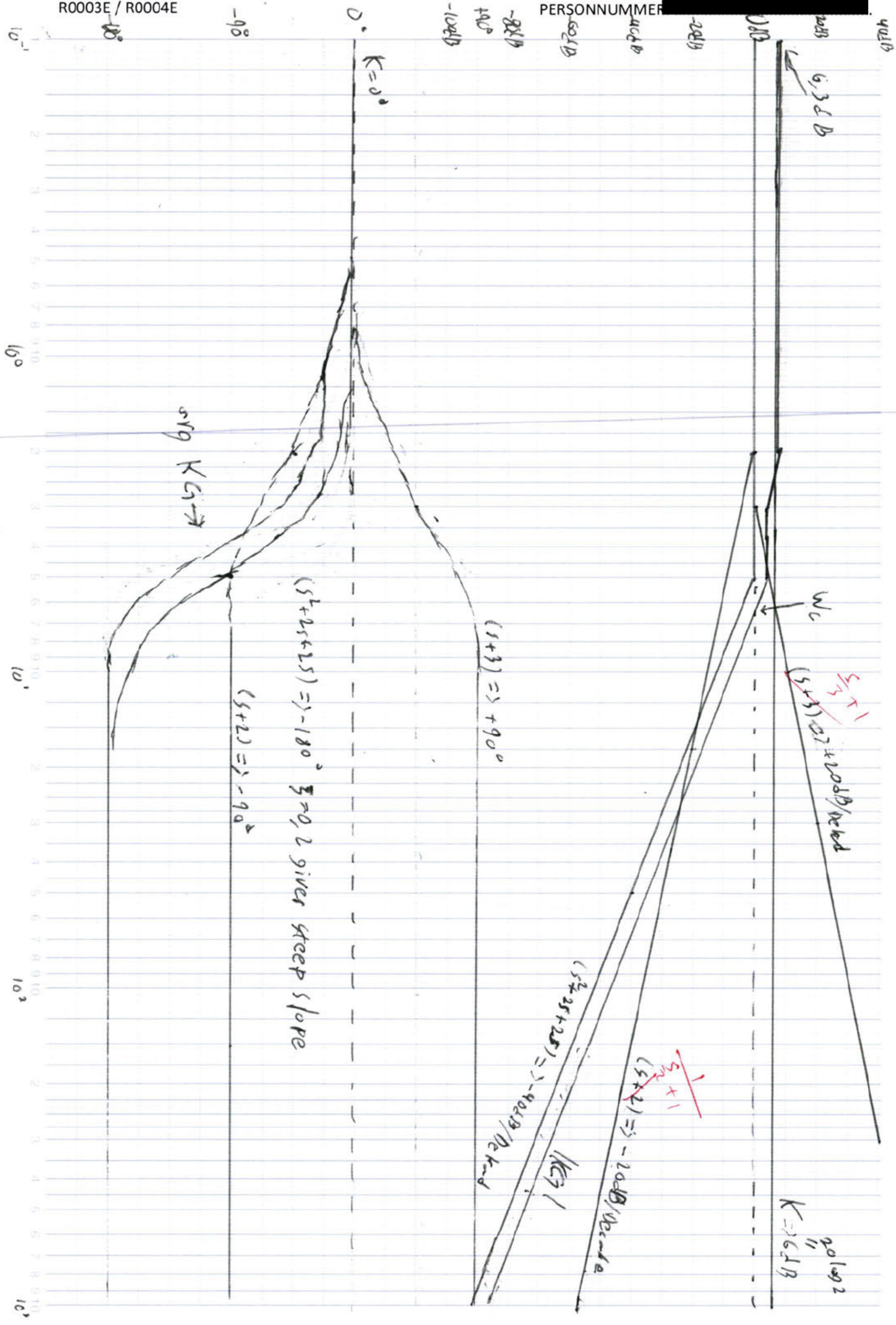
b. From Bode plot: The phase does not truly pass -180° so $GM = \infty$ If it did we would see GM from $1/|G(j\omega_0)|$ where ω_0 is the frequency the phase crosses -180°
My diagram gives PM of $\approx 15^\circ$ by checking margin between -180° and $\arg KG$ when $|KG|$ crosses 0dB.

c. The system is stable since both GM and PM are positive.

a.) 2,4
b 0.5
c 0.5

Sida nr. 6

R0003E / R0004E PERSONNUMMER [redacted]



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Namn: [redacted]

Personnummer: [redacted]

Sidnr: 7

Uppgift nr: 7

Poäng: 3.0

Lärarens anteckning:

a. $(K_p + K_d s)\,G(s)$ Assuming unitary feedback:

$$\frac{\frac{6(K_p+K_d s)}{s^2+2s+6}}{1+\frac{6(K_p+K_d s)}{s^2+2s+6}} = \frac{6(K_p+K_d s)}{s^2+2s+6+6(K_p+K_d s)} = \frac{6K_p+6K_d s}{s^2+(2+K_d)s+6K_p+6}$$

The characteristic equation of a transfer function is the denominator, in this case: $s^2+(2+K_d)s+6K_p+6$ ← Answer 1

b. 2nd order system: $s^2+2w_0\xi s+w_0^2$ From this we get 0.4

$$\begin{cases} w_0^2 = 6K_p+6 & (1) \\ 2w_0\xi = 2+K_d & (2) \end{cases} \quad (1):\ w_0=\sqrt{6K_p+6}\ (3) \quad (3) \text{ in } (2):\ 2\sqrt{6K_p+6}\,\xi = 2+6K_d \iff$$

$$\iff \xi = \frac{2+K_d}{2\sqrt{6K_p+6}} \quad \text{Answer: } \xi = \frac{2+K_d}{2\sqrt{6K_p+6}} \quad w_0=\sqrt{6K_p+6}$$ 1.3

c. $8=\sqrt{6K_p+6} \iff 6K_p+6=64 \iff 6K_p=58 \iff K_p=\frac{24}{3}$

$$0{,}8=\frac{2+K_d}{2\sqrt{6\cdot\frac{24}{3}+6}} \iff 0{,}8=\frac{2+6K_d}{16} \iff K_d=\cancel{10{,}8}$$

Answer: $K_p=\frac{24}{3}$ $K_d=10{,}8$

d. The PD controller (Proportional, Derivative) makes the system quicker and more accurate in its response to changes in reference values and disturbances by acting on a prediction of the systems future state (Following the derivative of the error value). 0.3

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Own notes

PID-regulator: u=K(e + 1/Ti*∫e(t)dt + Td*e´(t)), e = systemets fel, K,Ti,Td = valda konstanter

Tillståndsform (state-space): n x för n:te grad, x1 mostly = y, x2 = y'etc Derivera x1'... x' = Ax+Bu y = Cx+Du tot kvad mat

Linjärisering: **1)** Bestäm stationär punkt (x0, u0) x'0 = 0 ⇔ f(x0, u0) = 0 **2)** Taylorutveckla f och g kring (x0, u0): f(x, u) ≈ ∂f/∂x(x0,u0)(x-x0) + ∂f/∂u(x0, u0)(u-u0) | g(x,u) ≈ y0 + ∂g/∂x(x0, u0)(x-x0) + ∂g/∂u(x0, u0)(u-u0) | **3)** Inför nya variabler: Δx=x-x0 Δu=u-u0 Δy=y-y0 | **4)** Tillståndsekv. i nya var: Δ'x = x' - x'0 = x' = f(x,u) ≈ ∂f/∂x(x0,u0)Δx + ∂f/∂u(x0, u0)Δu = AΔx + BΔu | Δy = y - y0 = g(x, u) - y0 ≈ ∂g/∂x(x0, u0)Δx + ∂g/∂u(x0, u0)Δu = CΔx + DΔu **Om icke-skalär:** ex 2st: som ABCD men x är x-koordinat i mat. och f är y-koord. Så ∂f1/∂x2 i övre högra hörnet av A-matrisen. Och t.ex ∂g/∂x = (∂g/∂x1 ∂g/∂x2)

G(s) = Q(s)/P(s) Rötter till Q = nollställen, Rötter till P = Poler. Y(s) = G(s)U(s) PID ger Gr(s) = K(1 + 1/sTi + sTd) Enkel återkoppling: Y = (GrGp/(1+GrGp))R ← closed loop transfer with unitary feedback

Slutvärdesteoremet = Final Value Theorem och finns i FS. Samma med begynnelse. Statisk förstärkning = G(0) Första ordningens system: G(s) = K/(1+sT), pol i s = -1/T. Mindre T = snabbare stegsvar = pol längre från 0. Cos av vinkeln för ett system med komplexa poler ger relativ dämpning = relative damping.

Ökad frekvens ger oftast lägre förstärkning och sämre fasvridning. u(t) = K*sin(ωt) yt(t) = K|G|sin(ωt + φ) Förstärkning: |G| **Fasvridning**: arg(G) | G på formen A/B ger arg(G) = arg(A) - arg(B), arg(var) på formen arg(A*B) ger arg(var) = arg(A) + arg(B), för n som är heltal arg(x^n) = n*arg(x) | arg(x+iy) = arctan(y/x) för de enskilda argumenten arg(G(jω)) = arctan(Im(G(jω))/Re(G(jω))) | |G(jω)| = √(Re(G(jω))^2 + Im(G(jω))^2)

Bodediagram: G = G1G2G3 (dela för antal poler) log|G| = log|G1| + log|G2| + log|G3| Lutning = +-20dB/dekad täljare bryter upp, nämnare ner.
Typsystem: **G1,2)** Ks^n |G1,2| = Kω^n log|G1,2| = logK + nlogω arg(K(jω)^n) = nPi/2
G3) (1+sT): G(jω) = 1+jωT |1+jωT| = √(1^2 + omega^2T^2) låg frek: försumma ωT hög frek försumma 1 | arg(G3) = arctan(ωT/1), mellan 0-90° log(G3) = logω + logT Brytfrekvens = 1/T
G4) 1+2ζ(s/ω_n) + (s^2/ω_n^2) G(jω) = (1-ω^2/(ω_n^2)) + (j2ζ(ω/ω_n)) |G4(jω)| = √((1-ω^2/(ω_n^2))^2 + (j2ζ(ω/ω_n))^2) Låg frek ω<<ω_n ⇒ |G| = 1 Hög frek ω >> ω_n |G| = ω^2/ω_n^2 brytfrekvensen är ω=ω_n och bryter med 40db/dekad i.e lutning 2.
ζ = damping ratio, ω_n = **undampened natural frequency** **(20-21 under laplace i fs)**
G5) e^-sL G5(jω) = cos omegaL - jsinωL |G5| = 1 arg(G5) = -ωL. Bara fasvridning, 0 förstärkning.

Damping ratio ζ = (-ln(%OS / 100))/√(π^2 + ln^2(%OS / 100))
Phase margin φ_m = arctan(2ζ/√(-2ζ^2 + √1+4ζ^4))
Velocity error constant Kv = där ursprungslutningen skär 0dB. "the frequency where the extended magnitude plot prior corner frequency intersects with magnitude 0 dB. From the Bode plot, Kv = 300 rad/s"

Relation mellan in och utsignal för tidsfördröjning = y(t) = u(t-l). Laplace för tidsfördröjning = e^(-sL)*U(s) överföringsfunktion för denna = e^(-sL) = typsystem G5
dB till riktigt: 10^(dB/20) | Riktigt till dB: 20*log(riktigt)

Asymptotiskt stabilt: x(t) → 0 då t → ∞ för alla initialtillstånd. u(t) = 0 antaget för alla tre stabilitetsdefinitioner
Stabilt. Ett system är stabilt då x(t) ej konvergerar mot +-∞ när t → ∞ för alla initialtillstånd.
Instabilt: Ett system är instabilt då x(t) för *något* initialtillstånd konvergerar mot ∞.
Vi tittar på A-matrisens egenvärden: realdel < 0 för asym. stab. = 0 och unika imaginärdelar för stab. och > 0 för instabilitet. Titta på kar. pol. det(sI-A). Poler för G(s) = Egenvärden för A. Se FS för stabila kar. Ekv..

Slutna systemet G_0/(1+G_0) är asymptotiskt stabilt om punkten -1 ligger tills vänster sida i 'färdriktning' om nyquistkurvan (antagelsen görs att systemet möter kriterierna för vanlig stabilitet enligt kraven ovan).

Amplitudmarginal:Am = 1/(|G_0(jω_0)|), Am[2,6] ger bra beteende | Fasmarginal: φ_m = π+arg(G_0(jω_c)), φ_m[45, 60] ger bra beteene. | Dödtidsmarginal: Lm = φ_m/ω_c = fasmarginal/skärfrekvens I Bode Se ω_c på var amplitudkurvan passerar 1. Se fasmarginalen från frekvens ω_c:s vinkel till -180grader. ω_0 är var fasen skär -180grader. Se avståndet från amplitud 1 vid ω_0 för 1/A_m